

- Description

This module manages a smart power grid load balancing system, ensuring that:

1. **The total power demand in different grid sectors is continuously monitored to avoid overloads.**
2. **If power demand exceeds the safe threshold, non-essential loads are reduced or shifted to off-peak hours.**
3. **If power demand is low, surplus energy is stored or distributed efficiently to reduce waste.**
4. **A log of all energy usage, load adjustments, and efficiency optimizations is maintained for improved grid stability.**

This system helps prevent blackouts, optimize energy distribution, and improve overall power grid efficiency.

- Specification

module SmartPowerGridLoadBalancing

var currentLoad, maxLoad, minLoad: real

var powerSavingMode: boolean

var log: sequence of string

process updateLoad(changeAmount: real)

functional scenario form:

currentLoad $\geq$ minLoad && currentLoad = currentLoad + changeAmount, Power load balanced successfully.

||

currentLoad > maxLoad && powerSavingMode = true, log = log $\oplus$ ("Overload Warning: " + currentLoad + " MW, Activating Power-Saving Measures."), Overload warning triggered, power-saving measures activated.

||

currentLoad < minLoad && powerSavingMode = false, log = log $\oplus$ ("Low Demand Info: " + currentLoad + " MW, Storing Surplus Energy."), Low demand detected, surplus energy stored.

end process

end module

- Java Code

```
package code;
import java.util.*;
class SmartPowerGrid {
    private double maxLoad;
    private double minLoad;
    private double currentLoad;
    private boolean powerSavingMode;
    private List<String> log;
```

```

    public SmartPowerGrid(double maxLoad, double minLoad, double initialLoad) {
        this.maxLoad = maxLoad;
        this.minLoad = minLoad;
        this.currentLoad = initialLoad;
        this.powerSavingMode = false;
        this.log = new ArrayList<>();
    }
    public void updateLoad(double changeAmount) {
        currentLoad += changeAmount;
        manageLoadBalancing();
        String message = String.format("Current Power Load: %.2f MW", currentLoad);
        if (currentLoad > maxLoad) {
            message += " | WARNING: Overload Risk! Activating Power-Saving Measures.";
        } else if (currentLoad < minLoad) {
            message += " | INFO: Low Demand! Storing Surplus Energy.";
        } else {
            message += " | Load is Balanced.";
        }
        System.out.println(message);
        log.add(message);
    }
    private void manageLoadBalancing() {
        if (currentLoad > maxLoad) {
            powerSavingMode = true;
        } else {
            powerSavingMode = false;
        }
    }
    public void printLog() {
        System.out.println("\nPower Grid Load Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c54_SmartGridSystem {
    public static void main(String[] args) {
        SmartPowerGrid grid = new SmartPowerGrid(500.0, 100.0, 300.0);
        // testcase-VT:
        //     grid.updateLoad(150.0);
        //     grid.updateLoad(-200.0);
    }
}

```

```
//      grid.updateLoad(-180.0);
//      grid.updateLoad(250.0);
//      grid.updateLoad(250.0);
//      grid.printLog();

// testcase-FT:
//      grid.updateLoad(1.6279547056043695E293);
//      grid.updateLoad(1.4533186785200817E293);
//      grid.updateLoad(0.0);
//      grid.updateLoad(-1.7976931348623157E308);
//      grid.updateLoad(3.898125604559999E289);
//      grid.printLog();

// testcase-Z3:
//      grid.updateLoad(150);
//      grid.updateLoad(-200);
//      grid.updateLoad(-180);
//      grid.updateLoad(250);
//      grid.updateLoad(250);
//      grid.printLog();

// testcase-UVT:
grid.updateLoad(150.0);
grid.updateLoad(-200.0);
grid.updateLoad(-180.0);
grid.updateLoad(250.0);
grid.updateLoad(250.0);
grid.printLog();
}
}
```

● Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c54_SmartGridSystem.java	100.0 %	8	0	8
> c54_SmartGridSystem		0	0	0
> SmartPowerGrid	100.0 %	8	0	8

3. Vibration testing instruction coverage: 97.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	129	20,105	20,234
▼ src	0.6 %	129	20,105	20,234
▼ code	0.6 %	129	20,105	20,234
▼ c54_SmartGridSystem.java	97.7 %	129	3	132
> c54_SmartGridSystem	89.3 %	25	3	28
> SmartPowerGrid	100.0 %	104	0	104

4. Fuzz testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c54_SmartGridSystem.java	87.5 %	7	1	8
> c54_SmartGridSystem		0	0	0
> SmartPowerGrid	87.5 %	7	1	8

5. Fuzz testing instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	126	20,108	20,234
▼ src	0.6 %	126	20,108	20,234
▼ code	0.6 %	126	20,108	20,234
▼ c54_SmartGridSystem.java	95.5 %	126	6	132
> c54_SmartGridSystem	89.3 %	25	3	28
> SmartPowerGrid	97.1 %	101	3	104

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c54_SmartGridSystem.java	100.0 %	8	0	8
> c54_SmartGridSystem		0	0	0
> SmartPowerGrid	100.0 %	8	0	8

7. Z3 instruction coverage: 97.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	129	20,105	20,234
▼ src	0.6 %	129	20,105	20,234
▼ code	0.6 %	129	20,105	20,234
▼ c54_SmartGridSystem.java	97.7 %	129	3	132
> c54_SmartGridSystem	89.3 %	25	3	28
> SmartPowerGrid	100.0 %	104	0	104

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c54_SmartGridSystem.java	100.0 %	8	0	8
> c54_SmartGridSystem		0	0	0
> SmartPowerGrid	100.0 %	8	0	8

9. Unimproved V-method instruction coverage: 97.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	129	20,105	20,234
▼ src	0.6 %	129	20,105	20,234
▼ code	0.6 %	129	20,105	20,234
▼ c54_SmartGridSystem.java	97.7 %	129	3	132
> c54_SmartGridSystem	89.3 %	25	3	28
> SmartPowerGrid	100.0 %	104	0	104